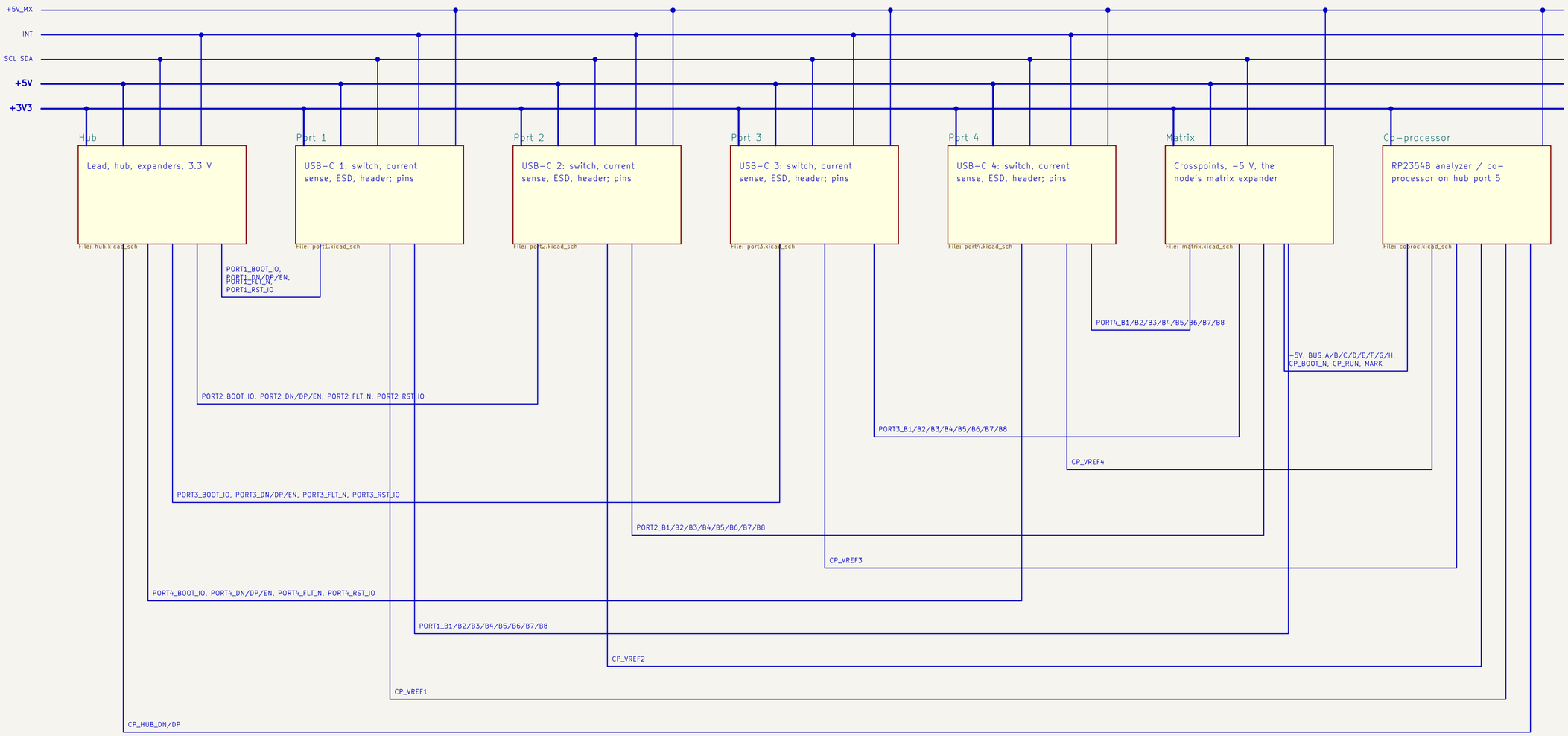



DATA ×4 v2.0 — USB2517I hub (7 ports), four USB-C (sub-slots A1,1-A1,4) each with a TPS2553-1, an INA226 and ESD, two TCA6408A for EN/FAULT/RESET/BOOT, an AP2114H 3.3 V; the PH 8 lead from the node's J7.
Pins: 32 on two CH446Q (±5 V). LSF0108 per port, an RP2354B analyzer on hub port 5, a TCA9534A for the matrix and the co-processor's RUN/BOOTSEL. Standing behind the port wall, one face assembled: v1 stays the other bays' board.
Lead pinout here: 1 INT · 2 SCL · 3 SDA · 4 GND · 5 D− · 6 D+ · 7 GND · 8 +5V (the node's J7 mirrored).
I²C: TCA6408A 0x20 / 0x21, INA226 0x41 / 0x43 / 0x44 / 0x45, TCA9534A 0x38; the node's INA226 0x40.
Assembled as DATA-4x-logic (as drawn) or DATA-4x (BOM/CPL '-base': the matrix, the co-processor and the ports' pin chains not fitted).

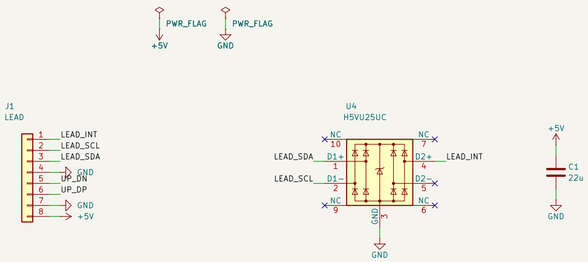
Generated from electronics/data-port-x4-v2/design.py — edit the Python, then regenerate
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, a logic an
**hw-testing-rack**

Sheet: /
File: data-port-x4-v2-6l.kicad_sch

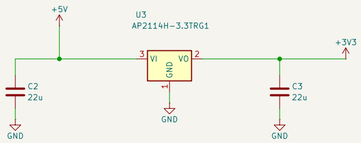
***Title: DATA ×4 v2 — Overview***

| Size: A2 | Date: | **Rev: 2.0** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 1/8 |

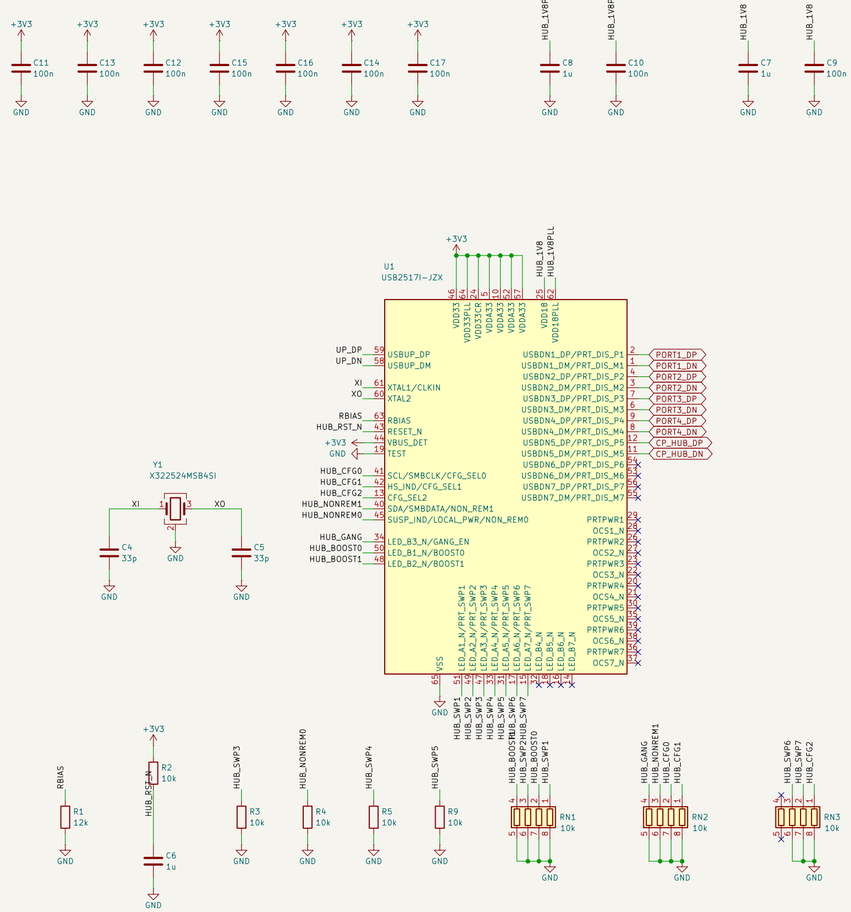
## Lead from the bay node's J7 (JST PH 8, vertical, on the rear face): SDA, SCL and INT through a H5VU25UC



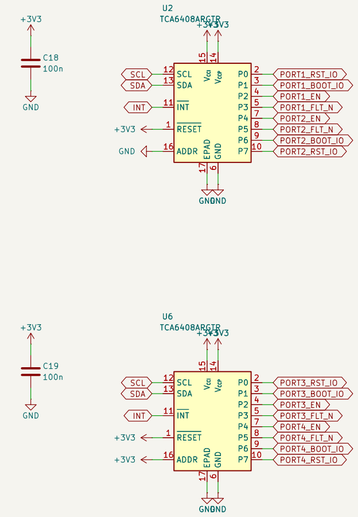

## 3.3 V for the hub, the expanders and the INA226s (AP2114H, SOT-223 on copper)



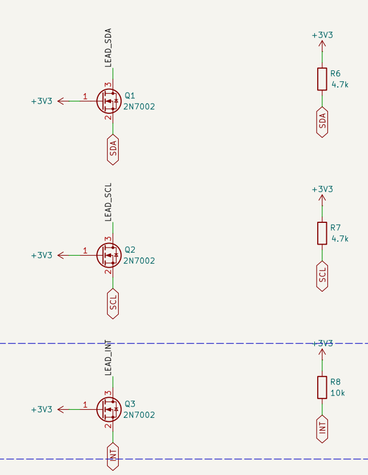

## USB2517I hub on its straps (CFG_SEL 000: internal default, straps on), 24 MHz crystal




## Two TCA6408A expanders, one per pair of ports: EN, FAULT_N, RESET, BOOT




## I²C isolation: open with DUT_5V off, so the node's pull-ups cannot feed this board




Hub straps: CFG_SEL[2:0] = 000 (internal default, strap options on). NON_REM 00, GANG_EN 0 (ports switched one by one: here by the expanders), BOOST 00, PRT_SWP 0000000: ports 6 and 7 declared but unused (cosmetic). On purpose: PRTPWR unused and OCS_N on their pull-ups (the node switches each port and reads its FAULT_N through the expanders); VBUS_DET on 3.3 V (self-powered hub).
Hub port n = USB-C n (busid 1-1.n) for n = 1-4: port 5 = the co-processor.
Expander pins are inputs at power-up; the EN pull-downs keep every port off until the node writes.
INT (J1.1 → the node's IO4): both expanders' INT and the four INA226 ALERTs, open drain, wired-OR, through Q3.
I²C and INT: the node's pull-ups on LEAD_*. 4.7k / 4.7k / 10k here: with DUT_5V off the chips here lose their registers; the node sets them up again after each DUT_5V on, and leaves this board's addresses alone while off.
J1 (through-hole) stands on the rear face: soldered by hand (the PCBA places the front face only).

Generated from electronics/data-port-x4-v2/design.py — edit the Python, then regenerate
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, a logic an

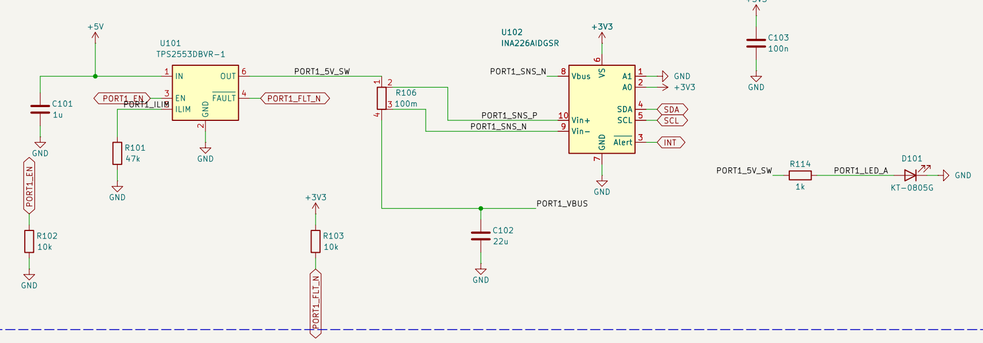
**hw-testing-rack**

Sheet: /Hub/
File: hub.kicad_sch

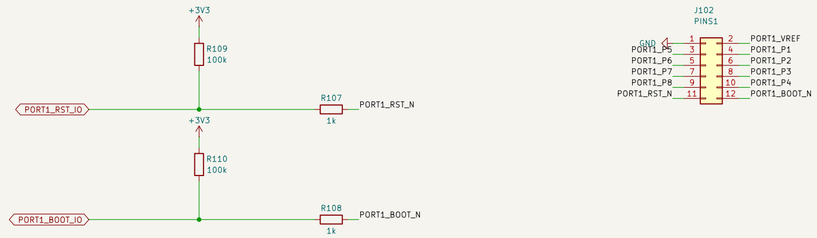
***Title: DATA x4 v2 — Lead, hub, expanders, 3.3 V***

| Size: A2 | Date: | **Rev: 2.0** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 2/8 |

## Port 1 power: +5V → TPS2553-1 → PORT1_5V_SW → 100 mΩ shunt → PORT1_VBUS, the INA226 across the shunt

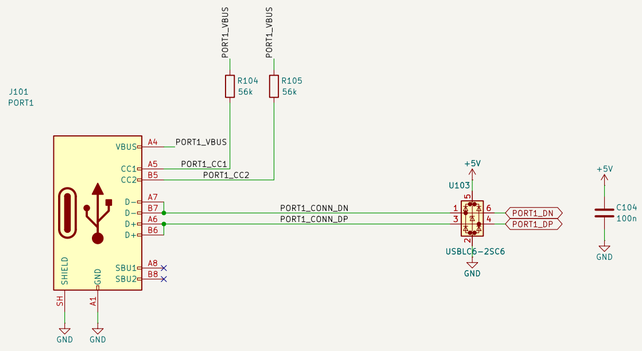



## Header 1 (2 × 6 through-hole): GND, VREF, P1-P8, RESET_N, BOOT_N; RESET/BOOT through 1k, their expander lines pulled up

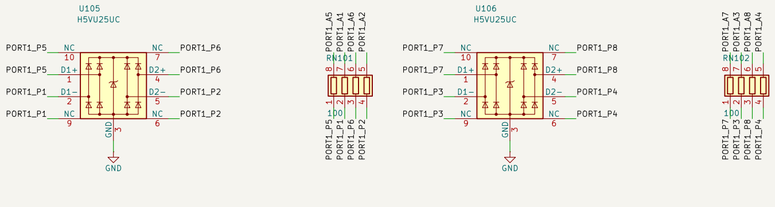



Port 1 = sub-slot A1.1, hub port 1, busid 1-1.1.
INA226 at 0x41: expander U2 (0x20).
ILIM 47k: 0.50-0.62 A; the −1 latches off after 7.5 ms in limit, FAULT_N flags it.
INA226 Vbus on the shunt's load-side sense pad (= VBUS).
EN pulled down: the port stays off until the node switches it on.
Pins: P1-P8 → crosspoint X lines (docs/electronics.md › DATA ×4 v2 map):
  the LSF0108 passes nothing while its VREF is at 0 V or unconnected (a board that is off).
VREF: the board's I/O supply, 0.95-3.6 V; a 5 V board with TTL inputs: its own 3.3 V pin.
Source only (Rp from the switched VBUS): a back-feeding source trips the TPS2553 (latched).
The DUT's RESET/BOOT pull-ups ≥ 4.7k: against 1k, the 1k in series stops at ≈ 1.65 V.
VBUS LED: on while the port's switch is on (5V_SW, before the shunt).

## USB-C 1 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)




## Pins 1: ESD in flow-through, 100 Ω, LSF0108 (A = the DUT's VREF, B = 3.3 V) to crosspoint U7's X lines




Generated from electronics/data-port-x4-v2/design.py — edit the Python, then regenerate.
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, a logic an

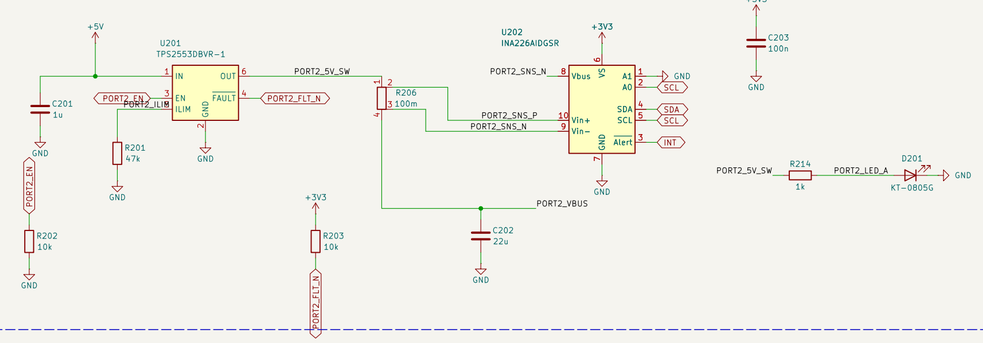
**hw-testing-rack**

Sheet: /Port 1/
File: port1.kicad_sch

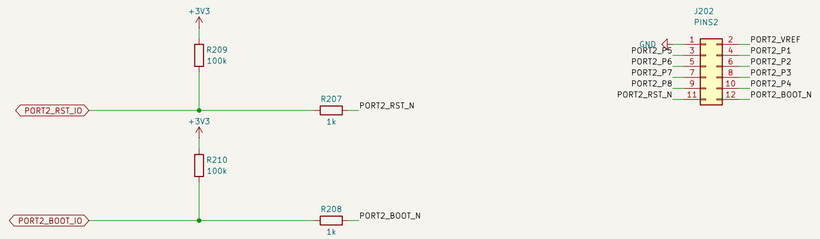
**Title: DATA ×4 v2 — USB-C 1: switch, current sense, ESD, header; pins**

| Size: A2 | Date: | Rev: 2.0 |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 3/8 |

## Port 2 power: +5V → TPS2553-1 → PORT2_5V_SW → 100 mΩ shunt → PORT2_VBUS, the INA226 across the shunt

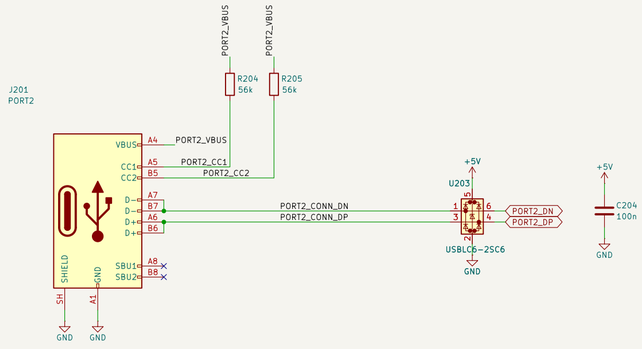



## Header 2 (2 × 6 through-hole): GND, VREF, P1-P8, RESET_N, BOOT_N; RESET/BOOT through 1k, their expander lines pulled up

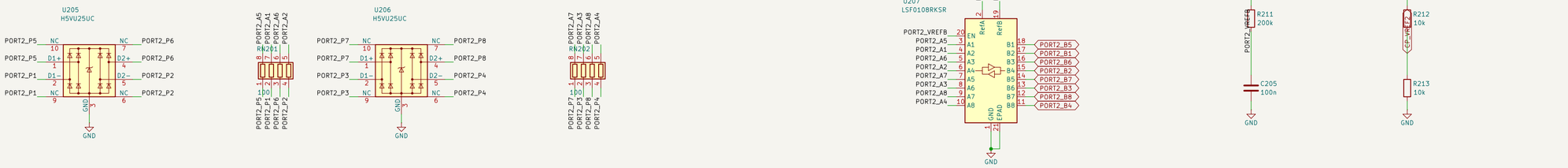



Port 2 = sub-slot A1.2, hub port 2, busid 1-1.2.
INA226 at 0x43; expander U2 (0x20).
ILIM 47k: 0.50-0.62 A; the ~1 latches off after 7.5 ms in limit, FAULT_N flags it.
INA226 Vbus on the shunt's load-side sense pad (= VBUS).
EN pulled down: the port stays off until the node switches it on.
Pins: P1-P8 → crosspoint X lines (docs/electronics.md › DATA ×4 v2 map):
  the LSF0108 passes nothing while its VREF is at 0 V or unconnected (a board that is off).
VREF: the board's I/O supply, 0.95-3.6 V; a 5 V board with TTL inputs: its own 3.3 V pin.
Source only (Rp from the switched VBUS): a back-feeding source trips the TPS2553 (latched).
The DUT's RESET/BOOT pull-ups ≥ 4.7k: against 1k, the 1k in series stops at ≈ 1.65 V.
VBUS LED: on while the port's switch is on (5V_SW, before the shunt).

## USB-C 2 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)




## Pins 2: ESD in flow-through, 100 Ω, LSF0108 (A = the DUT's VREF, B = 3.3 V) to crosspoint U7's X lines




Generated from electronics/data-port-x4-v2/design.py — edit the Python, then regenerate.
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, a logic an

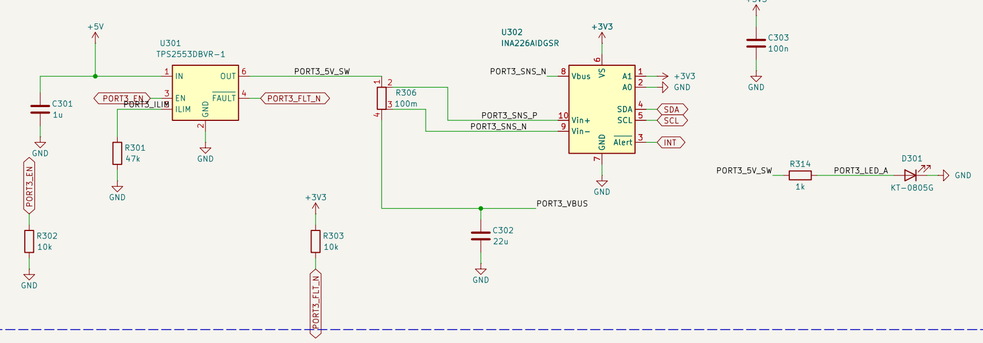
**hw-testing-rack**

Sheet: /Port 2/
File: port2.kicad_sch

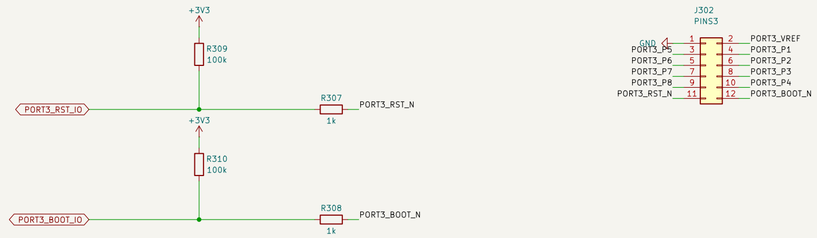
**Title: DATA ×4 v2 — USB-C 2: switch, current sense, ESD, header; pins**

| Size: A2 | Date: | Rev: 2.0 |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 4/8 |

## Port 3 power: +5V → TPS2553-1 → PORT3_5V_SW → 100 mΩ shunt → PORT3_VBUS, the INA226 across the shunt

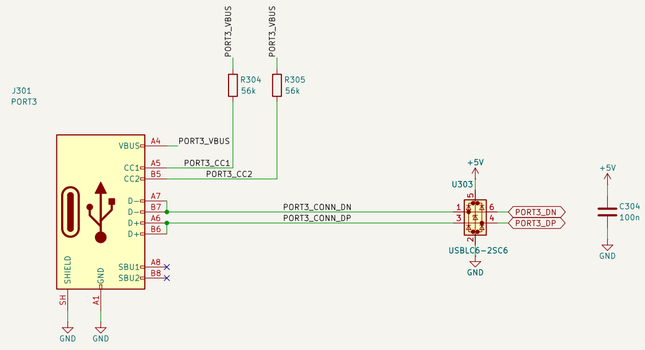



## Header 3 (2 × 6 through-hole): GND, VREF, P1-P8, RESET_N, BOOT_N; RESET/BOOT through 1k, their expander lines pulled up

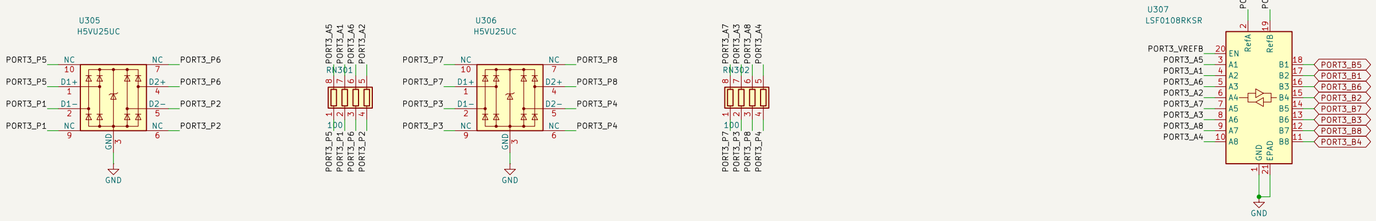



Port 3 = sub-slot A1.3, hub port 3, busid 1-1.3.
INA226 at 0x44: expander U6 (0x21).
ILIM 47k: 0.50-0.62 A; the ~1 latches off after 7.5 ms in limit, FAULT_N flags it.
INA226 Vbus on the shunt's load-side sense pad (= VBUS).
EN pulled down: the port stays off until the node switches it on.
Pins: P1-P8 → crosspoint X lines (docs/electronics.md › DATA ×4 v2 map):
  the LSF0108 passes nothing while its VREF is at 0 V or unconnected (a board that is off).
VREF: the board's I/O supply, 0.95-3.6 V; a 5 V board with TTL inputs: its own 3.3 V pin.
Source only (Rp from the switched VBUS): a back-feeding source trips the TPS2553 (latched).
The DUT's RESET/BOOT pull-ups ≥ 4.7k: against 1k, the 1k in series stops at ≈ 1.65 V.
VBUS LED: on while the port's switch is on (5V_SW, before the shunt).

## USB-C 3 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)




## Pins 3: ESD in flow-through, 100 Ω, LSF0108 (A = the DUT's VREF, B = 3.3 V) to crosspoint U8's X lines



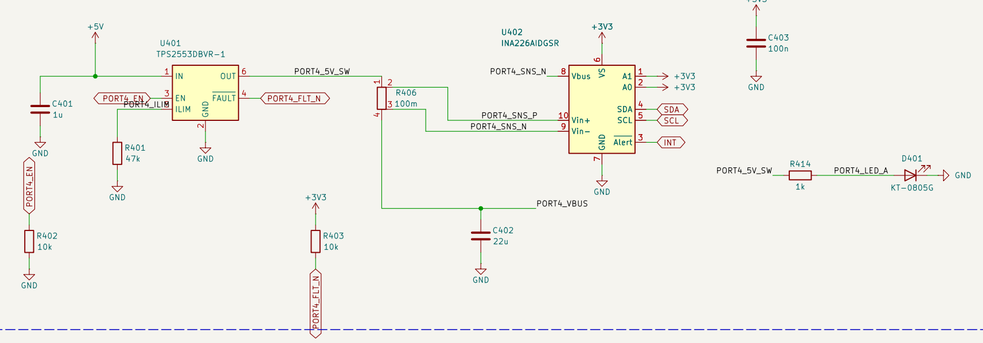

Generated from electronics/data-port-x4-v2/design.py — edit the Python, then regenerate.
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, a logic an
**hw-testing-rack**

Sheet: /Port 3/
File: port3.kicad_sch

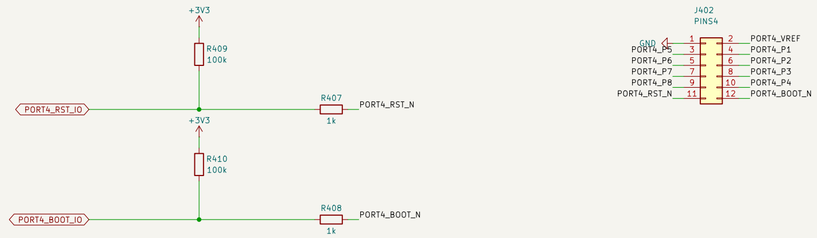
**Title: DATA ×4 v2 — USB-C 3: switch, current sense, ESD, header; pins**

| Size: A2 | Date: | **Rev: 2.0** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 5/8 |

## Port 4 power: +5V → TPS2553-1 → PORT4_5V_SW → 100 mΩ shunt → PORT4_VBUS, the INA226 across the shunt

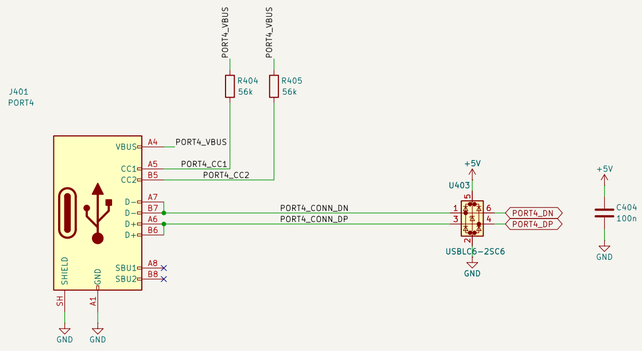



## Header 4 (2 × 6 through-hole): GND, VREF, P1-P8, RESET_N, BOOT_N; RESET/BOOT through 1k, their expander lines pulled up

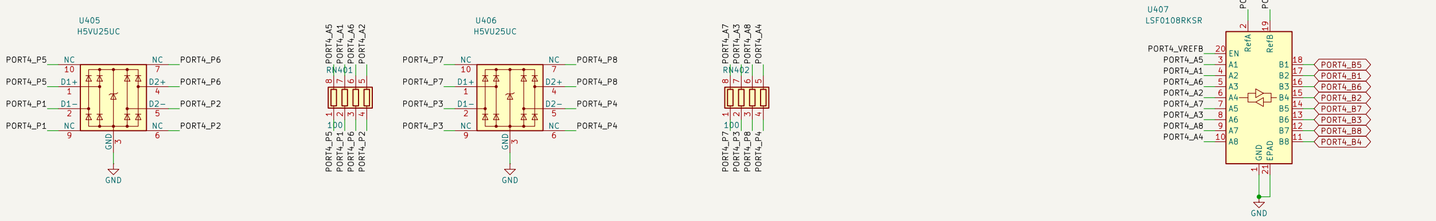



Port 4 = sub-slot A1.4, hub port 4, busid 1-1.4.
INA226 at 0x45; expander U6 (0x21).
ILIM 47k: 0.50-0.62 A; the −1 latches off after 7.5 ms in limit, FAULT_N flags it.
INA226 Vbus on the shunt's load-side sense pad (= VBUS).
EN pulled down: the port stays off until the node switches it on.
Pins: P1-P8 → crosspoint X lines (docs/electronics.md › DATA ×4 v2 map):
  the LSF0108 passes nothing while its VREF is at 0 V or unconnected (a board that is off).
VREF: the board's I/O supply, 0.95-3.6 V; a 5 V board with TTL inputs: its own 3.3 V pin.
Source only (Rp from the switched VBUS): a back-feeding source trips the TPS2553 (latched).
The DUT's RESET/BOOT pull-ups ≥ 4.7k: against 1k, the 1k in series stops at ≈ 1.65 V.
VBUS LED: on while the port's switch is on (5V_SW, before the shunt).

## USB-C 4 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)




## Pins 4: ESD in flow-through, 100 Ω, LSF0108 (A = the DUT's VREF, B = 3.3 V) to crosspoint U8's X lines




Generated from electronics/data-port-x4-v2/design.py — edit the Python, then regenerate.
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, a logic an

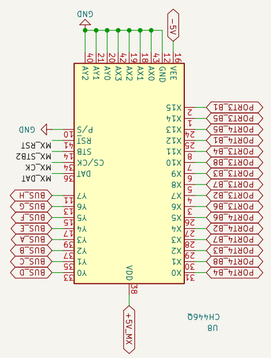
**hw-testing-rack**

Sheet: /Port 4/
File: port4.kicad_sch

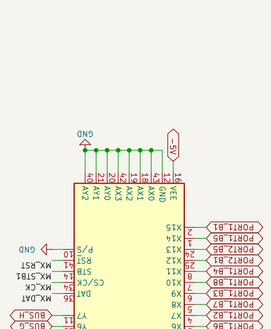
**Title: DATA ×4 v2 — USB-C 4: switch, current sense, ESD, header; pins**

| Size: A2 | Date: | Rev: 2.0 |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 6/8 |

## Crosspoint U8 (CH446Q at ±5 V, serial addressing): X = ports 3-4's pins, Y0-Y7 = buses DCBAEFGH




## Crosspoint U7 (CH446Q at ±5 V, serial addressing): X = ports 1-2's pins, Y0-Y7 = buses DCBAEFGH

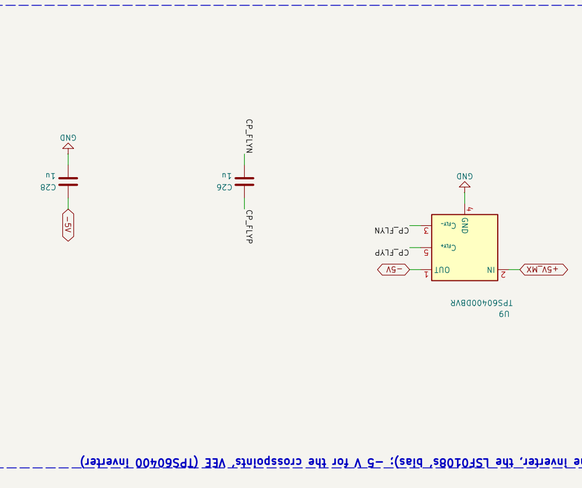



## The node's matrix expander (TCA9534, 0x38): the crosspoints and the co-processor's RUN / BOOTSEL

CH446Q serial mode (P/S low): 7 address bits on CK, data on DAT, latched by STB; RST high = all open.
TCA9534 0x38: P0 DAT, P1 CK, P2 STB, P3 STB, U8, P4 RST, P5 RUN, P6 BOOTSEL, P7 mark.
RUN and BOOTSEL are open drain emulated (output register 0, the pin switched to output).
±5 V on purpose: at a single 5 V R_on is 120-225 Ω, and a 1.8 V board's I²C low would fail.




## +5V_MX through a ferrite (the crosspoints, the inverter, the LSF0108' bias): -5 V for the crosspoints' VEE (TPS60400 inverter)

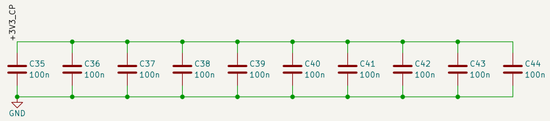


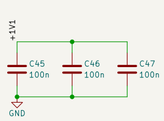

Generated from electronics/data-port-x4-v2/design.py — edit the Python, then regenerate.
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, a logic an

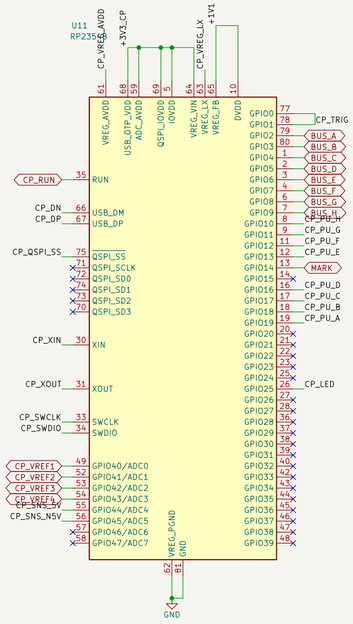
**Title: DATA ×4 v2 — Crosspoints, −5 V, the node's matrix expander**

| | |
|---|---|
| Sheet: /Matrix/ | |
| File: matrix.kicad_sch | |
| Size: A2 | Date: |
| KiCad E.D.A. 10.0.6 | Id: 7/8 |
| **hw-testing-rack** | **Rev: 2.0** |

## RP2354B, wired like a Pico 2 (gusmanb's PICO_2 UF2 runs as it is); its decoupling

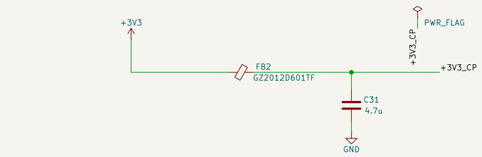


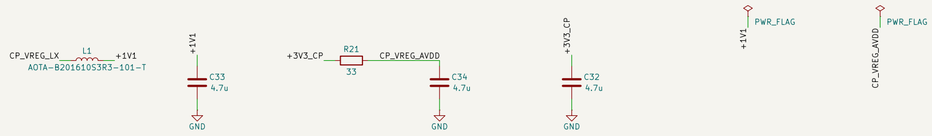



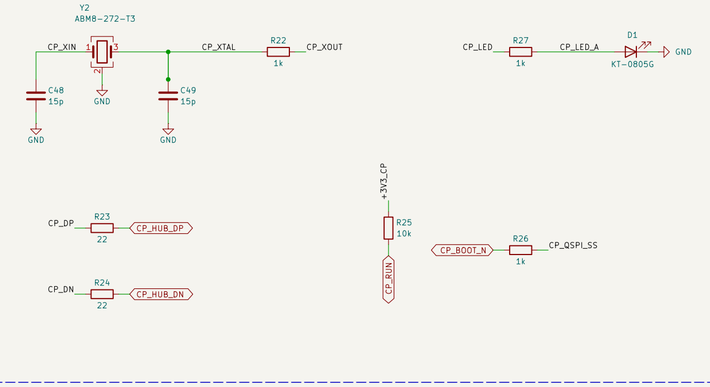

GPIO2-9 = buses A-H (gusmanb CH1-8, 8-bit mode); GPIO0 → GPIO1 bridged (its trigger path).
GPIO14 = the node's mark (TCA9534A P7); GPIO25 = LED; GPIO19…16, 13…10 = pull-up enables A…H (2.2k);
GPIO40-43 = VREF 1-4 / 2; GPIO44 = +5V × 47/147; GPIO45 = 3.3 V − (3.3 + 5) × 47/147 from −5 V.
RUN pulled up, BOOTSEL (QSPI_SS) through 1k; both held by the node (TCA9534A), open drain.
Core regulator: L1 polarity dot towards VREG_LX; C32, C33, C34 and L1 as close as the Pico 2's.

## Its 3.3 V from +3V3 through a ferrite, the core regulator as Raspberry Pi lays it out (3.3 µH, 3 × 4.7 µF, 33 Ω)

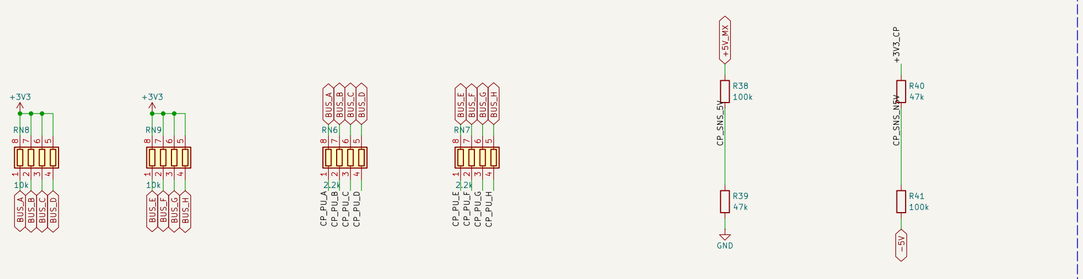



## 12 MHz crystal (ABM8-272-T3, 15 pF, 1k on XOUT), USB on hub port 5 (22 Ω), RUN, BOOTSEL, LED, SWD




## Bus pull-ups: 10k to 3.3 V always (the LSFs' B side), 2.2k switched by the co-processor (GPIO10-13, 16-19); the 15 V rails into the ADC




Generated from electronics/data-port-x4-v2/design.py — edit the Python, then regenerate.
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, a logic an

**hw-testing-rack**

Sheet: /Co-processor/
File: coproc.kicad_sch

***Title: DATA ×4 v2 — RP2354B analyzer / co-processor on hub port 5***

| Size: A2 | Date: | **Rev: 2.0** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 8/8 |